

Hamiltonian Graph-

If there exists a closed walk in the connected graph that visits every vertex of the graph exactly once.

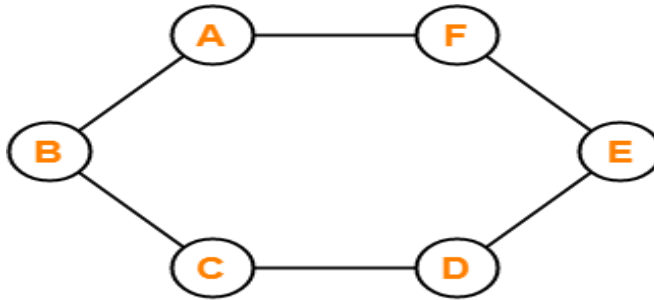
(except starting vertex) without repeating the edges,

then such a graph is called as a Hamiltonian graph.

“Or”

Any connected graph that contains a Hamiltonian circuit is called as a Hamiltonian Graph.

Hamiltonian Graph Example



Example of Hamiltonian Graph

Here,

This graph contains a closed walk ABCDEFA.

It visits every vertex of the graph exactly once except starting vertex.

The edges are not repeated during the walk.

Therefore, it is a Hamiltonian graph.

Alternatively, there exists a Hamiltonian circuit ABCDEFA in the above graph, therefore it is a Hamiltonian graph.

Hamiltonian Path

- If there exists a walk in the connected graph that visits every vertex of the graph exactly once without repeating the edges, then such a walk is called as a Hamiltonian path.

OR

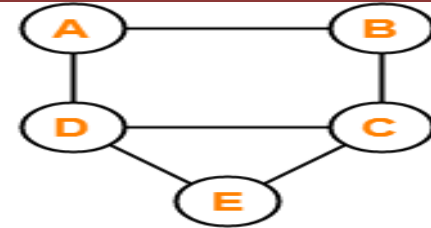
- If there exists a Path in the connected graph that contains all the vertices of the graph, then such a path is called as a Hamiltonian path.

NOTE

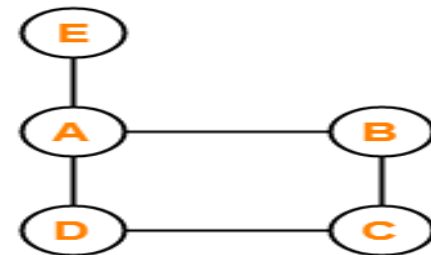
In Hamiltonian path, all the edges may or may not be covered but edges must not repeat.

Hamiltonian Path Examples

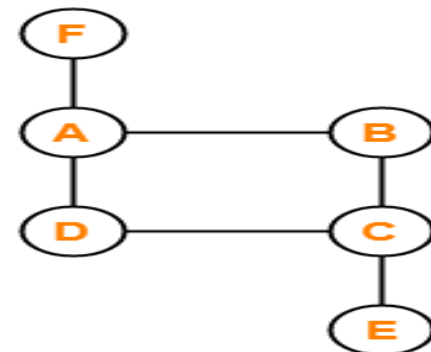
Hamiltonian Path Examples



Hamiltonian Path = ABCDE



Hamiltonian Path = EABCD



Hamiltonian Path Does Not Exist

Hamiltonian Circuit

- Hamiltonian circuit is also known as **Hamiltonian Cycle**.
- If there exists a walk in the connected graph that visits every vertex of the graph exactly once (except starting vertex) without repeating the edges and returns to the starting vertex, then such a walk is called as a Hamiltonian circuit.

Hamiltonian Circuit

OR

- If there exists a Cycle in the connected graph that contains all the vertices of the graph, then that cycle is called as a Hamiltonian circuit.

OR

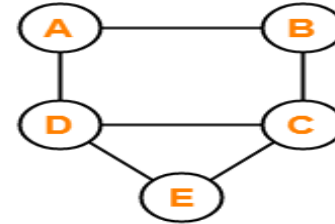
- A Hamiltonian path which starts and ends at the same vertex is called as a Hamiltonian circuit.

OR

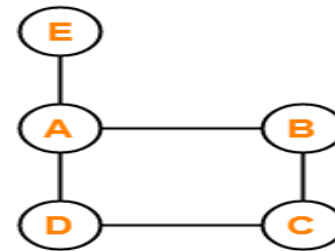
- A closed Hamiltonian path is called as a Hamiltonian circuit.

Hamiltonian Circuit Examples

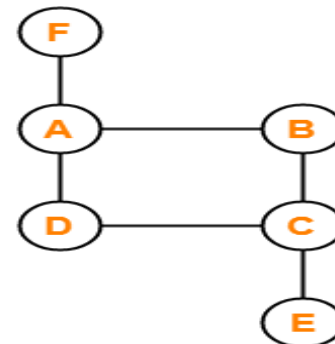
Hamiltonian Circuit Examples



Hamiltonian Circuit = ABCEDA



Hamiltonian Circuit Does Not Exist

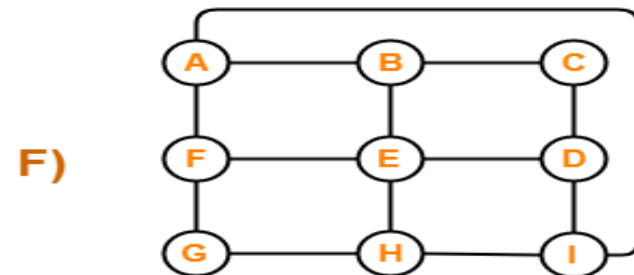
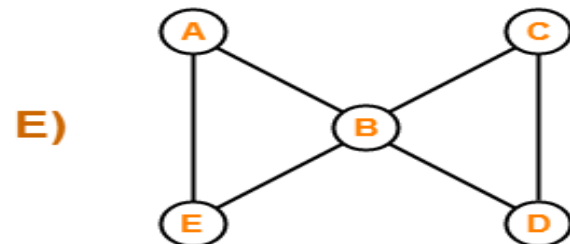
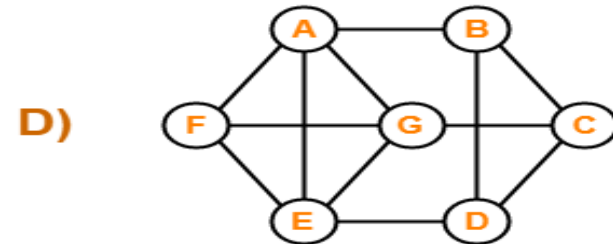
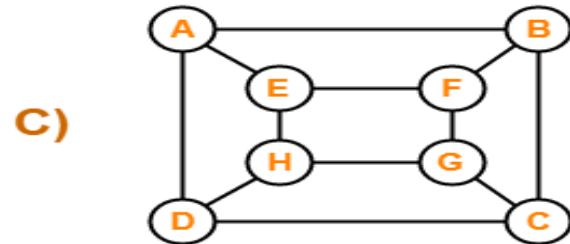
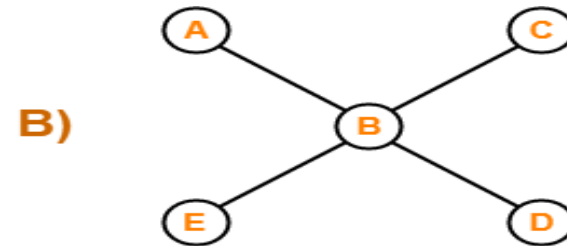
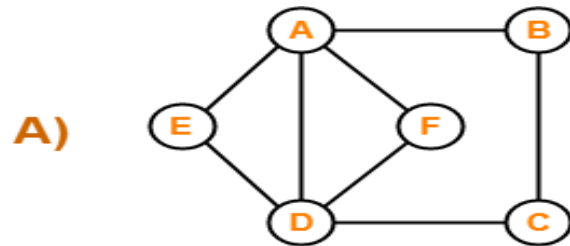


Hamiltonian Circuit Does Not Exist

Important Notes

- Any Hamiltonian circuit can be converted to a Hamiltonian path by removing one of its edges.
- Every graph that contains a Hamiltonian circuit also contains a Hamiltonian path but vice versa is not true.
- There may exist more than one Hamiltonian paths and Hamiltonian circuits in a graph.

Which of the following is / are Hamiltonian graphs?



Solution

A)

The graph neither contains a Hamiltonian path nor it contains a Hamiltonian circuit.

- Since graph does not contain a Hamiltonian circuit, therefore **It is not a Hamiltonian Graph.**

B)

- The graph neither contains a Hamiltonian path nor it contains a Hamiltonian circuit.
- Since graph does not contain a Hamiltonian circuit, therefore **It is not a Hamiltonian Graph.**

Solution

C)

- The graph contains both a Hamiltonian path (ABCDHGFE) and a Hamiltonian circuit (ABCDHGFEA).
- Since graph contains a Hamiltonian circuit, therefore **It is a Hamiltonian Graph.**

D)

- The graph contains both a Hamiltonian path (ABCDEFGF) and a Hamiltonian circuit (ABCDEFGFA).
- Since graph contains a Hamiltonian circuit, therefore **It is a Hamiltonian Graph.**

Solution

E)

- The graph neither contains a Hamiltonian path nor it contains a Hamiltonian circuit.
- Since graph does not contain a Hamiltonian circuit, therefore **It is not a Hamiltonian Graph.**

F)

- The graph contains both a Hamiltonian path (ABCDEFGHI) and a Hamiltonian circuit (ABCDEFGHIA)
- Since graph contains a Hamiltonian circuit, therefore **It is a Hamiltonian Graph.**